



LOCKDOWN CREW



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Classical Techniques



Substitution

Substitution ciphers replace letters with others, altering the plaintext's appearance significantly.



Transposition

Transposition ciphers rearrange the letters of the plaintext while retaining their identities.

History of Row Column Transposition Cipher

17th Century

1685

late 19th -Early
20th century

Modern Context

The concept of transposition ciphers, including row and column arrangements started to appear in European cryptography. Samuel Morland, an English mathematician, introduced early forms of columnar transposition

John Falconer, a British cryptographer, published a detailed explanation of columnar transposition ciphers. At this time, these methods were mainly used by scholars and government clerks in the UK.

became widely known among European cryptographers. It was used for military and diplomatic communications because it was simple to implement yet difficult for casual interceptors to decode.

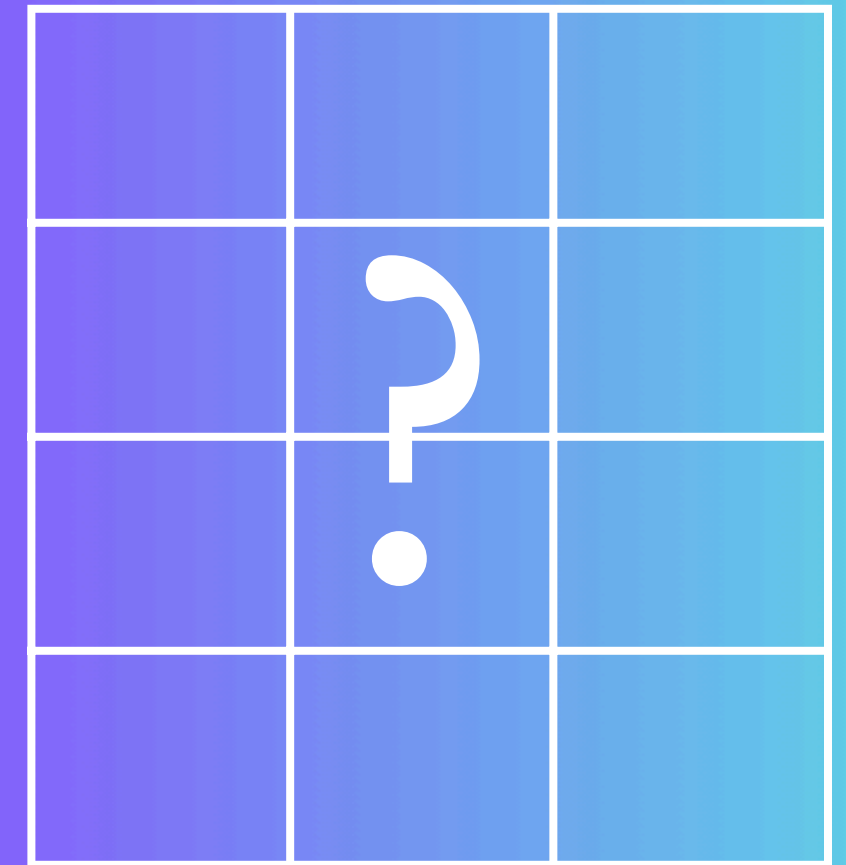
While replaced by more advanced cryptographic algorithms, row-column transposition ciphers are still taught today to illustrate classical cryptography techniques.

Encryption Process

Steps for efficient message encryption

To build a matrix

1. Choose a key that determines the order of columns.
2. num of columns = length of key ,
num of rows = length of plain text / length of key .
3. Write the plaintext into rows .
4. If the last row is incomplete add padding (usually X).
5. Put the number-based key over the matrix columns to simplify arranging it.
6. Read the columns in the key order to generate the ciphertext.



Encryption Example #1

Plaintext: ATTACKATDAWN

Key: 3 1 4 2

Number Of Columns: 4

Column order (1 → 2 → 3 → 4):

Order 1 = Column 2 → T K A

Order 2 = Column 4 → A T N

Order 3 = Column 1 → A C D

Order 4 = Column 3 → T A W

3	1	4	2
A	T	T	A
C	K	A	T
D	A	W	N

Ciphertext: TKAATNACDTAW

Encryption Example #2

Plaintext: MEETMEHERE

Key: 3 1 2

Columns = 3 $\rightarrow$ Rows = 4 (with padding)

M	E	E
T	M	E
H	E	R
E	X	X

Column order:

Order 1 $\rightarrow$ Column 2 $\rightarrow$ E M E X

Order 2 $\rightarrow$ Column 3 $\rightarrow$ E E R X

Order 3 $\rightarrow$ Column 1 $\rightarrow$ M T H E

Ciphertext: EMEXEERXMTHE

SPECIAL CASE >



SPECIAL CASE

Plaintext: encryption

Key: plain

Number Of Columns: 5

5	3	1	2	4
e	n	c	r	y
p	t	i	o	n

PUT THE KEY ON ALPHABETIC ORDER AS FOLLOW:

Key: 5 3 1 2 4

Ciphertext: cironytnep

Decryption Process

Reverse the encryption steps accurately.

1. Determine number of rows = $\text{ciphertext_length} \div \text{number_of_columns}$.
2. Split the ciphertext into segments according to the key order.
3. Rebuild the matrix column-by-column.
4. Read the plaintext row-by-row.
5. Remove any padding characters (e.g., X).



Decryption Example #1

Ciphertext: TKAATNACDTAW

Key: [3 1 4 2] → Columns = 4 → Rows = $12/4 = 3$

Recovered Plaintext: ATTACKATDAWN

Split ciphertext:

Order 1 → Column 2: T K A

Order 2 → Column 4: A T N

Order 3 → Column 1: A C D

Order 4 → Column 3: T A W

3	1	4	2
A	T	T	A
C	K	A	T
D	A	W	N

Decryption Example #2

Split ciphertext:
Order 1 → Column 2: E M E X
Order 2 → Column 3: E E R X
Order 3 → Column 1: M T H E

Ciphertext: EMEXEERXMTHE

Key: [3 1 2] → Columns = 3 → Rows = $12/3 = 4$

Recovered Plaintext : MEETMEHERE

3	1	2
M	E	E
T	M	E
H	E	R
E	X	X

GUI Implementation

row column transposition cipher - lock down crew

lock down crew

row column transposition cipher

enter text:

zeinabdiab

permutation key (e.g. 3 1 4 2):

3 1 4 2

encrypt

decrypt

output:

EBBNIZAAID

SPECIAL CASES

*the key is not
permutation*

row column transposition cipher - lock down crew

lock down crew

row column transposition cipher

enter text:

zeinabdiab

permutation key (e.g. 3 1 4 2):

3 1

encrypt decrypt

output:

EBBNIZAAID

Error

key is not a valid permutation

OK

PYTHON CODE

Using Tkinter

```
zeinabsecpy.py • x.html
E: > zeinabsecpy.py
1 import tkinter as tk
2 from tkinter import messagebox
3
4 def ispermutationzad(keylistzad):
5     n = len(keylistzad)
6     return sorted(keylistzad) == list(range(1, n+1))
7
8 def encryptzad(textzad, keylistzad):
9     textzad = textzad.replace(" ", "").upper()
10    if not ispermutationzad(keylistzad):
11        raise ValueError("key is not a valid permutation")
12
13    n = len(keylistzad)
14    rows = (len(textzad) + n - 1) // n
15
16    matrixzad = [[''] * n for _ in range(rows)]
17    idx = 0
18    for r in range(rows):
19        for c in range(n):
20            if idx < len(textzad):
21                matrixzad[r][c] = textzad[idx]
22                idx += 1
23
24    reading_order = sorted(range(n), key=lambda c: keylistzad[c])
25
26    ciphertextzad = ""
27    for c in reading_order:
28        for r in range(rows):
29            if matrixzad[r][c]:
30                ciphertextzad += matrixzad[r][c]
31    return ciphertextzad
32
33 def decryptzad(cipherzad, keylistzad):
```



W

O

R

D

P

U

Z

Z

L

E



guess the key from ciphertext and plaintext



ENCRYPTION PUZZLE – GUESS THE KEY

WHICH PERMUTATION KEY WAS USED?

A) 3 1 4 2

B) 4 3 1 2

C) 2 4 1 3.

Original phrase:

"DUMTUMSALIMEEN"

Encrypted output:

"MSMTAEUMINDULE"

